Explainable Political Bias Classification and Mitigation Using Transformer-Based Deep Learning

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***Abstract*—Abstract— The rapid growth of online news platforms and digital media has significantly increased the spread of politically biased content and ideologically polarized narratives. Detecting political bias in news articles has become an important challenge in Natural Language Processing (NLP) because political language often contains subtle contextual framing and semantically overlapping ideological patterns. Traditional machine learning approaches relying on handcrafted features are limited in capturing such contextual relationships effectively. This work presents an explainable multi-class political bias classification and mitigation framework using transformer-based deep learning models. The proposed system classifies political news articles into five ideological categories: Center, Leaning-Left, Left, Leaning-Right, and Right. The methodology includes dataset preprocessing, tokenization, class balancing, transformer fine-tuning, explainability analysis, and mitigation-oriented adaptation. Multiple transformer architectures, including DistilBERT, ALBERT, RoBERTa, and BART combined with Low-Rank Adaptation (LoRA), were experimentally evaluated for bias classification performance. Among all models, RoBERTa achieved the best overall performance with approximately 76–77% accuracy, a weighted F1-score of nearly 0.77, and strong ROC-AUC performance across ideological categories. To improve interpretability, SHAP-based explainable AI analysis was integrated into the framework to identify influential political entities, contextual phrases, and semantic patterns contributing to ideological predictions. In addition, the BART + LoRA mitigation framework reduced computational complexity and enabled parameter-efficient transformer adaptation for scalable deployment. Experimental results demonstrate that combining transformer-based contextual learning with explainable AI techniques provides an effective and interpretable framework for political bias classification and mitigation. The proposed work contributes an integrated transformer-based political bias analysis framework combining classification, explainability, and mitigation within a unified NLP system.**

***Index Terms*—Political Bias Classification, Bias Mitigation, Explainable Artificial Intelligence (XAI), Natural Language Processing (NLP), Transformer Models, RoBERTa, SHAP, Multi-Class Classification, BART, LoRA, Political News Analysis, Deep Learning, Text Classification, Ideological Detection, Explainable NLP**

I. Introduction

The rapid growth of digital media platforms and online news consumption has significantly transformed the way political information is created, distributed, and interpreted. Millions of users rely on online news portals, social media platforms, blogs, and digital journalism sources for political updates and public discourse. However, the increasing volume of online political content has also led to the widespread circulation of ideologically biased narratives, partisan framing, and politically polarized reporting. News organizations often present political events using selective language, contextual framing, and emotionally influenced narratives that reflect specific ideological viewpoints. Such biased information can influence public opinion, intensify political polarization, and contribute to the spread of misinformation. Consequently, automatic political bias classification and mitigation have become important research problems in Natural Language Processing (NLP), Artificial Intelligence (AI), and computational social science.

Political bias classification refers to the task of automatically identifying the ideological orientation of textual content. Unlike conventional sentiment analysis, political bias detection is more complex because ideological patterns are often subtle, context-dependent, and semantically overlapping. Political articles may contain indirect framing, selective emphasis, contextual persuasion, and narrative structures that collectively indicate ideological leaning. Traditional machine learning approaches such as Support Vector Machines (SVM), Naive Bayes, Logistic Regression, and Random Forest classifiers have previously been used for political text classification. Although these methods achieved moderate performance, they relied heavily on handcrafted linguistic features and shallow text representations, limiting their ability to capture contextual semantics and long-range dependencies within political language.

Recent advancements in transformer-based deep learning architectures have significantly improved NLP performance across various text classification tasks. Models such as DistilBERT, ALBERT, BERT, RoBERTa, and BART utilize self-attention mechanisms to learn contextual relationships between words and phrases, enabling deeper semantic understanding of textual data. Transformer architectures have demonstrated superior performance in sentiment analysis, misinformation detection, hate speech detection, and political ideology classification because they effectively capture contextual information from large-scale corpora. Among these architectures, RoBERTa has shown particularly strong contextual represen-

tation learning capability due to its optimized pretraining strategy and dynamic masking mechanism.

Despite these advancements, several important challenges remain in existing political bias detection systems. First, many studies primarily focus on predictive accuracy while providing limited interpretability and transparency. Transformer models often behave as black-box systems, making it difficult to understand why a specific ideological prediction was generated. In politically sensitive applications, explainability is essential to ensure trustworthiness and transparency. Second, many existing works focus only on binary political classification, which oversimplifies real-world ideological diversity. Multi-class ideological classification involving nuanced categories such as Left, Leaning-Left, Center, Leaning-Right, and Right remains comparatively underexplored. Third, large transformer architectures require substantial computational resources and GPU memory, limiting scalability and deployment feasibility in resource-constrained environments.

To address these limitations, recent explainable AI research has emphasized the importance of integrating interpretability techniques into deep learning systems. Explainable Artificial Intelligence (XAI) methods such as SHAP (SHapley Additive Explanations) provide token-level and global feature contribution analysis, enabling better understanding of transformer decision-making behavior. Explainability techniques improve transparency by identifying influential political entities, contextual phrases, and semantic patterns contributing to ideological predictions. In addition, parameter-efficient transformer adaptation techniques such as Low-Rank Adaptation (LoRA) have emerged as effective strategies for reducing computational overhead during transformer fine-tuning.

Motivated by these challenges, the proposed work presents an explainable multi-class political bias classification and mitigation framework using transformer-based deep learning models. Multiple transformer architectures, including DistilBERT, ALBERT, RoBERTa, and BART combined with Low-Rank Adaptation (LoRA), were experimentally explored and comparatively evaluated. The proposed system classifies political news articles into five ideological categories: Center, Leaning-Left, Left, Leaning-Right, and Right. Among all experimented models, RoBERTa achieved the strongest overall performance because of its superior contextual representation learning capability. To improve interpretability and transparency, SHAP-based explainability analysis was integrated into the framework to identify influential political entities, contextual semantic relationships, and ideological framing patterns contributing to predictions. In addition, a mitigation-oriented BART + LoRA framework was explored to reduce computational complexity and improve parameter-efficient transformer adaptation for scalable deployment.

The major contributions of the proposed work are summarized as follows:

- Development of a transformer-based multi-class political bias classification framework for ideological news analysis.
- Comparative evaluation of DistilBERT, ALBERT, RoBERTa, and BART + LoRA architectures for political bias classification.
- Integration of weighted learning and class balancing strategies to improve minority class prediction performance.
- Implementation of SHAP-based explainable AI analysis for token-level and global ideological interpretation.
- Exploration of parameter-efficient transformer mitigation using LoRA adaptation for scalable deployment.
- Comprehensive evaluation using weighted F1-score, ROC-AUC analysis, confusion matrices, and explainability-based interpretation.

The remainder of this paper is organized as follows. Section II discusses related work and literature associated with political bias detection, transformer architectures, explainable AI, and bias mitigation techniques. Section III explains the proposed methodology, including preprocessing, model building, transformer fine-tuning, explainability analysis, and mitigation framework. Section IV presents the experimental results, comparative analysis, visualization-based interpretation, and error analysis. Finally, Section V concludes the work and discusses future research directions.

II. Related Work / Literature Review

Political bias detection has emerged as an important research area in Natural Language Processing (NLP), computational journalism, and Explainable Artificial Intelligence (XAI). Early approaches primarily relied on traditional machine learning algorithms combined with handcrafted textual features for ideological classification. Although these methods achieved moderate performance, they faced limitations in contextual understanding and semantic representation learning. Recent advancements in transformer-based deep learning architectures have significantly improved contextual semantic analysis in political text classification. At the same time, explainability and mitigation-oriented transformer adaptation have become increasingly important for improving transparency, fairness, and deployment scalability.

Initial political bias classification systems utilized conventional machine learning algorithms such as Support Vector Machines (SVM), Logistic Regression, Naive Bayes, Decision Trees, and Random Forest classifiers. These models commonly relied on handcrafted features including Term Frequency-Inverse Document Frequency (TF-IDF), Bag-of-Words representations, lexical patterns, sentiment scores, and topic modeling techniques. While such approaches provided computational simplicity and acceptable baseline performance, they struggled to capture long-range contextual dependencies and semantically overlapping ideological narratives present in political news articles. Political language often contains implicit framing, contextual persuasion, and nuanced ideological expressions that cannot be effectively represented using shallow textual features alone.

With the advancement of deep learning, researchers began exploring neural architectures such as Convolutional Neural

Networks (CNNs), Recurrent Neural Networks (RNNs), and Long Short-Term Memory (LSTM) networks for political text analysis and sentiment classification. CNN-based models improved local feature extraction capability by identifying influential textual patterns and n-gram structures. Similarly, LSTM architectures improved sequential contextual learning by preserving temporal relationships between words and sentences. Although these models achieved better semantic understanding compared to traditional machine learning approaches, they still faced limitations in capturing global contextual dependencies within long-form political narratives.

The introduction of transformer architectures significantly improved contextual representation learning in NLP applications. Vaswani et al. [1] introduced the transformer architecture using self-attention mechanisms capable of learning contextual relationships across entire textual sequences. Building upon this foundation, Devlin et al. [2] proposed BERT (Bidirectional Encoder Representations from Transformers), which introduced bidirectional contextual embeddings and significantly improved semantic understanding across multiple NLP tasks. BERT demonstrated strong performance in sentiment analysis, document classification, question answering, and misinformation detection.

Subsequent transformer architectures further improved performance and efficiency. DistilBERT [4] reduced transformer complexity using knowledge distillation while preserving most contextual representation capability of BERT. ALBERT [5] improved parameter efficiency using embedding factorization and parameter-sharing mechanisms, thereby reducing memory consumption and computational complexity. RoBERTa [3] optimized transformer pretraining through dynamic masking, larger pretraining corpora, and improved training strategies, resulting in stronger contextual semantic learning and improved classification accuracy.

Recent studies have increasingly explored transformer architectures for political bias detection and ideological classification. Transformer-based models demonstrated strong capability in identifying subtle ideological framing, contextual political narratives, and semantically overlapping political categories. However, many existing studies primarily focused on binary classification tasks such as liberal versus conservative or left versus right classification. Real-world political ideology exists on a broader semantic spectrum involving nuanced categories such as Center, Leaning-Left, Left, Leaning-Right, and Right. Multi-class political bias classification therefore remains a comparatively challenging and underexplored problem.

Another major limitation of existing transformer-based political classification systems is the lack of explainability and transparency. Transformer architectures often behave as black-box models, making it difficult to understand why certain ideological predictions are generated. In politically sensitive applications, interpretability is essential because classification outcomes may influence public trust, media analysis, and social perception. To address this challenge, researchers began integrating Explainable Artificial Intelligence (XAI) techniques into NLP systems.

Among explainability approaches, SHAP (SHapley Additive Explanations) proposed by Lundberg and Lee [8] became one of the most widely adopted frameworks for interpreting machine learning and transformer predictions. SHAP provides token-level and global feature contribution analysis by measuring the importance of input features for model decisions. In political text analysis, SHAP enables identification of influential political entities, ideological phrases, policy-related terminology, and contextual framing patterns contributing to classification outcomes. Recent studies demonstrated that explainability techniques improve transparency and trustworthiness in transformer-based NLP systems.

In addition to explainability, recent research has also focused on improving computational efficiency and scalability of large transformer architectures. Fine-tuning large transformer models often requires substantial GPU memory and computational resources, limiting deployment feasibility in practical environments. To address this challenge, parameter-efficient fine-tuning methods such as Low-Rank Adaptation (LoRA) were introduced. Hu et al. [7] proposed LoRA as a lightweight transformer adaptation strategy that injects trainable low-rank matrices into transformer attention layers while freezing most pretrained parameters. LoRA significantly reduces trainable parameter complexity while preserving contextual learning capability.

Recent research also explored transformer-based mitigation techniques for reducing ideological bias and generating more neutralized textual outputs. Sequence-to-sequence transformer architectures such as BART [6] demonstrated strong capability in text generation, summarization, rewriting, and contextual transformation tasks. Combining BART with LoRA adaptation enables mitigation-oriented transformer fine-tuning with lower computational overhead and improved scalability.

Although existing research has achieved considerable progress in political NLP systems, several limitations still remain. Many studies primarily emphasize predictive performance while providing limited explainability and interpretability. Several approaches focus only on binary ideological classification instead of nuanced multi-class analysis. In addition, computational complexity and scalability remain major challenges for large transformer architectures. Furthermore, relatively few studies integrate explainability, mitigation, and parameter-efficient adaptation within a unified framework.

The proposed work addresses these research gaps by developing an explainable multi-class political bias classification and mitigation framework using transformer-based deep learning architectures. Unlike existing approaches that focus solely on classification accuracy, the proposed framework integrates contextual transformer learning, SHAP-based explainability analysis, weighted learning for class imbalance handling, and mitigation-oriented LoRA adaptation within a unified NLP system. The framework performs ideological classification across five political categories while also improving transparency, interpretability, and computational efficiency. This combination contributes toward the development of more

trustworthy, scalable, and explainable political bias analysis systems.

III. Dataset / Data Description

The dataset used in this work consists of political news articles collected from multiple online news platforms representing different ideological viewpoints. The dataset was stored in CSV format and processed using the Pandas library for preprocessing, transformer tokenization, and model training. The collected data primarily focuses on political news reporting and ideological framing patterns commonly observed in digital journalism platforms.

The dataset contains approximately 5,800–6,000 political news articles after preprocessing and cleaning operations. Each record contains both textual and metadata-based attributes associated with political news reporting. The dataset includes long-form textual political content, making it suitable for transformer-based contextual semantic learning and ideological classification tasks.

A. Dataset Features

The original dataset contains multiple attributes related to political news articles. The important features present in the dataset include:

- URL
- Topic
- Date
- Title
- Site
- Bias
- Page_Text

Among these features, the primary input feature used for transformer training was:

`page_text`

which contains the complete textual content of political news articles.

The target output feature used for supervised learning was:

`bias`

which represents the ideological orientation associated with each article.

B. Dataset Type

The dataset primarily contains unstructured textual data consisting of long-form political news articles. Since the problem focuses on ideological text classification, the dataset belongs to the domain of Natural Language Processing (NLP) and multi-class text classification.

The political articles include:

- Political discussions
- Election coverage
- Government policies
- Ideological narratives
- Political opinions
- Policy debates
- Media framing patterns

The textual content contains contextual semantic relationships, ideological framing structures, and politically influenced narratives that require deep contextual representation learning for effective classification.

C. Ideological Classes

The dataset contains five ideological categories representing different political viewpoints:

TABLE I
Political Bias Categories

Class	Description
Center	Neutral or balanced political reporting
Leaning-Left	Moderately left-oriented political framing
Left	Strongly left-oriented political ideology
Leaning-Right	Moderately right-oriented political framing
Right	Strongly right-oriented political ideology

These ideological categories form a multi-class political classification problem where neighboring ideological groups often share semantically overlapping contextual patterns.

D. Dataset Challenges

The political bias dataset presents several challenges for machine learning and transformer-based classification systems.

One of the primary challenges is contextual ideological overlap. Political categories such as Left and Leaning-Left or Right and Leaning-Right frequently share similar vocabulary, policy discussions, and contextual framing patterns. This semantic similarity makes precise ideological separation difficult.

Another major challenge is class imbalance. Some ideological categories contain significantly more samples than others, potentially biasing the model toward dominant classes during training. To address this issue, weighted loss balancing strategies were incorporated during transformer fine-tuning.

The dataset also contains long-form political narratives involving quotations, mixed viewpoints, and contextual persuasion techniques. Such complex textual structures increase semantic ambiguity and require advanced contextual representation learning mechanisms.

Additionally, political news articles often contain emotionally influenced language, indirect framing, and selective emphasis, making ideological classification more difficult than conventional sentiment analysis tasks.

E. Final Prepared Dataset

After preprocessing and cleaning operations, the final prepared dataset included:

- Cleaned political news articles
- Encoded ideological labels
- Tokenized transformer inputs
- Attention masks
- Balanced training configuration

The processed dataset was then used for transformer-based fine-tuning using DistilBERT, ALBERT, RoBERTa, and BART

+ LoRA architectures for political bias classification and mitigation analysis.

IV. Methodology / Proposed Work

The proposed methodology focuses on developing an explainable multi-class political bias classification and mitigation framework using transformer-based deep learning models and Explainable Artificial Intelligence (XAI) techniques. The overall workflow consists of dataset preprocessing, feature preparation, transformer-based model training, explainability analysis, mitigation-oriented adaptation, and performance evaluation. Initially, the political news dataset is cleaned and transformed into a suitable format for transformer learning. Multiple transformer architectures, including DistilBERT, ALBERT, RoBERTa, and BART combined with Low-Rank Adaptation (LoRA), are then fine-tuned for multi-class ideological classification. To improve learning stability and handle dataset imbalance, weighted loss balancing and early stopping strategies are incorporated during training. After model evaluation, SHAP-based explainability analysis is performed to identify influential political entities, contextual phrases, and semantic patterns contributing to predictions. Finally, parameter-efficient mitigation using LoRA adaptation is explored to reduce computational complexity and improve scalability for practical deployment.

A. Overall System Architecture

The proposed framework follows a sequential workflow for political bias classification and mitigation. The overall pipeline consists of:

1) Dataset Collection and Loading
2) Data Preprocessing and Cleaning
3) Label Encoding and Dataset Splitting
4) Transformer Tokenization
5) Model Fine-Tuning
6) Class Balancing and Optimization
7) Performance Evaluation
8) SHAP-Based Explainability Analysis
9) BART + LoRA Mitigation Framework

The system architecture integrates both classification and explainability components within a unified transformer-based NLP framework.

B. Data Preprocessing

The preprocessing stage prepares the political news dataset for transformer training. Initially, the dataset is loaded using the Pandas library, followed by removal of invalid or noisy samples.

The preprocessing pipeline includes:

- Removal of missing values
- Removal of duplicate articles
- Elimination of empty textual entries
- Basic text normalization
- Conversion of labels into categorical format

Minimal preprocessing was intentionally adopted because transformer architectures preserve contextual semantics more effectively when raw textual structure is maintained.

The primary feature used for training is:

`page_text`

while the target output label is:

`bias`

representing the ideological orientation of each article.

C. Label Encoding and Dataset Splitting

The ideological categories were converted into numerical labels for supervised learning.

The encoding scheme used was:

TABLE II
Political Bias Label Encoding

Political Bias	Encoded Label
Center	0
Leaning-Left	1
Leaning-Right	2
Left	3
Right	4

After preprocessing, the dataset was divided into:

- Training Set (70%)
- Validation Set (15%)
- Testing Set (15%)

A stratified splitting strategy was applied to preserve class distribution consistency across all subsets.

D. Transformer Tokenization

The cleaned political news articles were tokenized using pretrained transformer tokenizers corresponding to each transformer architecture.

The tokenization process included:

- Token splitting
- Attention mask generation
- Padding
- Truncation

A maximum sequence length of 256 tokens was selected to balance contextual coverage and computational efficiency.

The tokenization process generated:

- Input IDs
- Attention Masks

which were used as inputs for transformer fine-tuning.

E. Transformer Models Used

Multiple transformer architectures were experimentally explored and comparatively evaluated for political bias classification.

1) DistilBERT: DistilBERT was used as a lightweight transformer baseline because of its reduced parameter count and faster inference capability. The model provided lower computational complexity but demonstrated weaker contextual discrimination for semantically overlapping ideological categories.

2) ALBERT: ALBERT was explored as a parameter-efficient transformer architecture utilizing parameter sharing and reduced embedding complexity. Although ALBERT improved memory efficiency, it struggled to consistently separate nuanced ideological classes.

3) RoBERTa: RoBERTa was selected as the primary classification model because of its superior contextual representation learning and optimized pretraining strategy. The model demonstrated the strongest performance across all evaluation metrics and effectively captured ideological semantics within political text.

4) BART + LoRA Mitigation Model: A mitigation-oriented BART + LoRA framework was explored to reduce computational overhead and improve parameter-efficient transformer adaptation.

LoRA introduces trainable low-rank matrices into transformer attention layers while freezing most pretrained parameters. This significantly reduces trainable parameter count and memory usage while preserving contextual learning capability.

The mitigation framework improves:

- Computational efficiency
- Scalability
- Deployment feasibility

for large transformer systems.

F. Class Balancing Strategy

The dataset contained imbalanced ideological distributions, which could potentially bias the model toward dominant classes.

To address this issue, weighted class balancing was incorporated during training using weighted cross-entropy loss.

Class weights were calculated inversely proportional to class frequency:

- Minority classes received higher weights
- Dominant classes received lower weights

This balancing strategy improved minority class learning and enhanced weighted F1-score performance.

G. Model Training Strategy

The transformer models were fine-tuned using supervised learning on the political news dataset.

The training configuration included:

TABLE III
TRAINING CONFIGURATION

Hyperparameter	Configuration
Optimizer	AdamW
Maximum Epochs	40
Early Stopping Patience	4
Maximum Sequence Length	256
Loss Function	Weighted Cross-Entropy
Evaluation Metric	Weighted F1-Score

Training was performed on GPU hardware to accelerate transformer optimization.

Weighted F1-score was selected as the primary monitoring metric because the dataset contains multiple imbalanced ideological categories.

H. Early Stopping and Regularization

To prevent overfitting, early stopping was integrated during training.

The weighted F1-score on the validation dataset was monitored after each epoch. Training automatically stopped when validation performance failed to improve for consecutive epochs defined by the patience value.

This strategy improved:

- Model generalization
- Training stability
- Computational efficiency

I. Performance Evaluation

The trained transformer models were evaluated using multiple quantitative and visualization-based metrics.

The evaluation metrics included:

- Accuracy
- Precision
- Recall
- Weighted F1-Score
- Macro F1-Score
- ROC-AUC

Visualization-based analysis included:

- Confusion Matrices
- ROC Curves
- Precision-Recall Curves
- Confidence Distribution Analysis

These metrics provided comprehensive analysis of classification quality, class balance, and semantic discrimination capability.

J. SHAP-Based Explainability Analysis

To improve interpretability and transparency, SHAP (SHapley Additive Explanations) was integrated into the framework.

SHAP was used to analyze:

- Token-level importance
- Contextual contribution
- Ideological influence
- Semantic feature relevance

The explainability framework included:

1) Local Explanation: Analyzing individual article predictions by identifying influential political entities and contextual phrases responsible for ideological classification.

2) Global Explanation: Aggregating feature contributions across multiple articles to identify dominant ideological patterns learned by the transformer model.

SHAP visualizations included:

- Feature importance plots
- Beeswarm plots
- Token contribution analysis
- Global ideological interpretation

The explainability analysis demonstrated that the transformer models relied on contextual political semantics rather than isolated keyword matching.

K. Mitigation-Oriented LoRA Adaptation

The proposed work also explored mitigation-oriented parameter-efficient adaptation using LoRA.

Instead of updating all transformer parameters during fine-tuning, LoRA updates only lightweight trainable matrices inserted into transformer attention layers.

This approach provided several advantages:

- Reduced GPU memory usage
- Faster fine-tuning
- Lower computational overhead
- Efficient deployment capability

The BART + LoRA mitigation framework demonstrated that parameter-efficient adaptation can preserve competitive transformer performance while improving scalability and deployment feasibility.

L. Final Proposed Framework

The final proposed framework integrates:

- Transformer-based multi-class political bias classification
- Weighted learning and class balancing
- SHAP-based explainability analysis
- LoRA-based mitigation and parameter-efficient adaptation

within a unified NLP pipeline.

The framework combines:

- Contextual semantic learning
- Explainability
- Interpretability
- Computational efficiency

to create a scalable and trustworthy political bias classification and mitigation system.

V. Methodology / Proposed Work

The proposed methodology focuses on developing an explainable multi-class political bias classification and mitigation framework using transformer-based deep learning models and Explainable Artificial Intelligence (XAI) techniques. The overall workflow consists of dataset preprocessing, feature preparation, transformer-based model training, explainability analysis, mitigation-oriented adaptation, and performance evaluation. Initially, the political news dataset is cleaned and transformed into a suitable format for transformer learning. Multiple transformer architectures, including DistilBERT, ALBERT, RoBERTa, and BART combined with Low-Rank Adaptation (LoRA), are then fine-tuned for multi-class ideological classification. To improve learning stability and handle dataset imbalance, weighted loss balancing and early stopping strategies are incorporated during training. After model evaluation, SHAP-based explainability analysis is performed to identify influential political entities, contextual phrases, and semantic patterns contributing to predictions. Finally, parameter-efficient mitigation using LoRA adaptation is explored to reduce computational complexity and improve scalability for practical deployment.

A. Overall System Architecture

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9) BART + LoRA Mitigation Framework

The system architecture integrates both classification and explainability components within a unified transformer-based NLP framework.

B. Data Preprocessing

The preprocessing stage prepares the political news dataset for transformer training. Initially, the dataset is loaded using the Pandas library, followed by removal of invalid or noisy samples.

The preprocessing pipeline includes:

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4) BART + LoRA Mitigation Model: A mitigation-oriented BART + LoRA framework was explored to reduce computational overhead and improve parameter-efficient transformer adaptation.

LoRA introduces trainable low-rank matrices into transformer attention layers while freezing most pretrained parameters. This significantly reduces trainable parameter count and memory usage while preserving contextual learning capability.

The mitigation framework improves:

- Computational efficiency
- Scalability
- Deployment feasibility

for large transformer systems.

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The proposed work also explored mitigation-oriented parameter-efficient adaptation using LoRA.

Instead of updating all transformer parameters during fine-tuning, LoRA updates only lightweight trainable matrices inserted into transformer attention layers.

This approach provided several advantages:

- Reduced GPU memory usage
- Faster fine-tuning
- Lower computational overhead
- Efficient deployment capability

The BART + LoRA mitigation framework demonstrated that parameter-efficient adaptation can preserve competitive transformer performance while improving scalability and deployment feasibility.

L. Final Proposed Framework

The final proposed framework integrates:

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- SHAP-based explainability analysis
- LoRA-based mitigation and parameter-efficient adaptation

within a unified NLP pipeline.

The framework combines:

- Contextual semantic learning
- Explainability
- Interpretability
- Computational efficiency

to create a scalable and trustworthy political bias classification and mitigation system.

VI. Methodology / Proposed Work

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A. Overall System Architecture

The proposed framework follows a sequential workflow for political bias classification and mitigation. The overall pipeline consists of:

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2) Data Preprocessing and Cleaning
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4) Transformer Tokenization
5) Model Fine-Tuning
6) Class Balancing and Optimization
7) Performance Evaluation
8) SHAP-Based Explainability Analysis
9) BART + LoRA Mitigation Framework

The system architecture integrates both classification and explainability components within a unified transformer-based NLP framework.

B. Data Preprocessing

The preprocessing stage prepares the political news dataset for transformer training. Initially, the dataset is loaded using the Pandas library, followed by removal of invalid or noisy samples.

The preprocessing pipeline includes:

- Removal of missing values
- Removal of duplicate articles
- Elimination of empty textual entries
- Basic text normalization
- Conversion of labels into categorical format

Minimal preprocessing was intentionally adopted because transformer architectures preserve contextual semantics more effectively when raw textual structure is maintained.

The primary feature used for training is:

`page_text`

while the target output label is:

`bias`

representing the ideological orientation of each article.

C. Label Encoding and Dataset Splitting

The ideological categories were converted into numerical labels for supervised learning.

The encoding scheme used was:

TABLE VI
POLITICAL BIAS LABEL ENCODING

Political Bias	Encoded Label
Center	0
Leaning-Left	1
Leaning-Right	2
Left	3
Right	4

After preprocessing, the dataset was divided into:

- Training Set (70%)
- Validation Set (15%)
- Testing Set (15%)

A stratified splitting strategy was applied to preserve class distribution consistency across all subsets.

D. Transformer Tokenization

The cleaned political news articles were tokenized using pretrained transformer tokenizers corresponding to each transformer architecture.

The tokenization process included:

- Token splitting
- Attention mask generation
- Padding
- Truncation

A maximum sequence length of 256 tokens was selected to balance contextual coverage and computational efficiency.

The tokenization process generated:

- Input IDs
- Attention Masks

which were used as inputs for transformer fine-tuning.

E. Transformer Models Used

Multiple transformer architectures were experimentally explored and comparatively evaluated for political bias classification.

1) DistilBERT: DistilBERT was used as a lightweight transformer baseline because of its reduced parameter count and faster inference capability. The model provided lower computational complexity but demonstrated weaker contextual discrimination for semantically overlapping ideological categories.

2) ALBERT: ALBERT was explored as a parameter-efficient transformer architecture utilizing parameter sharing and reduced embedding complexity. Although ALBERT improved memory efficiency, it struggled to consistently separate nuanced ideological classes.

3) RoBERTa: RoBERTa was selected as the primary classification model because of its superior contextual representation learning and optimized pretraining strategy. The model demonstrated the strongest performance across all evaluation metrics and effectively captured ideological semantics within political text.

4) BART + LoRA Mitigation Model: A mitigation-oriented BART + LoRA framework was explored to reduce computational overhead and improve parameter-efficient transformer adaptation.

LoRA introduces trainable low-rank matrices into transformer attention layers while freezing most pretrained parameters. This significantly reduces trainable parameter count and memory usage while preserving contextual learning capability.

The mitigation framework improves:

- Computational efficiency
- Scalability
- Deployment feasibility

for large transformer systems.

F. Class Balancing Strategy

The dataset contained imbalanced ideological distributions, which could potentially bias the model toward dominant classes.

To address this issue, weighted class balancing was incorporated during training using weighted cross-entropy loss.

Class weights were calculated inversely proportional to class frequency:

- Minority classes received higher weights
- Dominant classes received lower weights

This balancing strategy improved minority class learning and enhanced weighted F1-score performance.

G. Model Training Strategy

The transformer models were fine-tuned using supervised learning on the political news dataset.

The training configuration included:

TABLE VII
TRAINING CONFIGURATION

Hyperparameter	Configuration
Optimizer	AdamW
Maximum Epochs	40
Early Stopping Patience	4
Maximum Sequence Length	256
Loss Function	Weighted Cross-Entropy
Evaluation Metric	Weighted F1-Score

Training was performed on GPU hardware to accelerate transformer optimization.

Weighted F1-score was selected as the primary monitoring metric because the dataset contains multiple imbalanced ideological categories.

H. Early Stopping and Regularization

To prevent overfitting, early stopping was integrated during training.

The weighted F1-score on the validation dataset was monitored after each epoch. Training automatically stopped when validation performance failed to improve for consecutive epochs defined by the patience value.

This strategy improved:

- Model generalization
- Training stability
- Computational efficiency

I. Performance Evaluation

The trained transformer models were evaluated using multiple quantitative and visualization-based metrics.

The evaluation metrics included:

- Accuracy
- Precision
- Recall
- Weighted F1-Score
- Macro F1-Score
- ROC-AUC

Visualization-based analysis included:

- Confusion Matrices
- ROC Curves
- Precision-Recall Curves
- Confidence Distribution Analysis

These metrics provided comprehensive analysis of classification quality, class balance, and semantic discrimination capability.

J. SHAP-Based Explainability Analysis

To improve interpretability and transparency, SHAP (SHapley Additive Explanations) was integrated into the framework.

SHAP was used to analyze:

- Token-level importance
- Contextual contribution
- Ideological influence
- Semantic feature relevance

The explainability framework included:

1) Local Explanation: Analyzing individual article predictions by identifying influential political entities and contextual phrases responsible for ideological classification.

2) Global Explanation: Aggregating feature contributions across multiple articles to identify dominant ideological patterns learned by the transformer model.

SHAP visualizations included:

- Feature importance plots
- Beeswarm plots
- Token contribution analysis
- Global ideological interpretation

The explainability analysis demonstrated that the transformer models relied on contextual political semantics rather than isolated keyword matching.

K. Mitigation-Oriented LoRA Adaptation

The proposed work also explored mitigation-oriented parameter-efficient adaptation using LoRA.

Instead of updating all transformer parameters during fine-tuning, LoRA updates only lightweight trainable matrices inserted into transformer attention layers.

This approach provided several advantages:

- Reduced GPU memory usage
- Faster fine-tuning
- Lower computational overhead
- Efficient deployment capability

The BART + LoRA mitigation framework demonstrated that parameter-efficient adaptation can preserve competitive transformer performance while improving scalability and deployment feasibility.

L. Final Proposed Framework

The final proposed framework integrates:

- Transformer-based multi-class political bias classification
- Weighted learning and class balancing
- SHAP-based explainability analysis
- LoRA-based mitigation and parameter-efficient adaptation

within a unified NLP pipeline.

The framework combines:

- Contextual semantic learning
- Explainability
- Interpretability
- Computational efficiency

to create a scalable and trustworthy political bias classification and mitigation system.

VII. EVALUATION METRICS

The performance of the proposed transformer-based political bias classification framework was evaluated using multiple quantitative and visualization-based metrics. Since the dataset contains multiple ideological classes with imbalanced class distributions, relying on a single evaluation metric is insufficient for comprehensive analysis. Therefore, several complementary metrics were used to evaluate classification

quality, contextual semantic discrimination capability, minority class learning performance, and overall model robustness.

The evaluation process focused on measuring both predictive performance and class-wise discrimination capability across different ideological categories. In addition, visualization-based evaluation techniques were used to analyze prediction behavior, confidence distribution, and explainability characteristics of transformer models.

A. Accuracy

Accuracy measures the overall proportion of correctly classified samples among all predictions generated by the model. It provides a general understanding of model performance across the complete dataset.

The accuracy metric is defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

where:

- TP represents True Positives
- TN represents True Negatives
- FP represents False Positives
- FN represents False Negatives

Accuracy was used to compare the overall prediction capability of DistilBERT, ALBERT, RoBERTa, and BART + LoRA models. However, since the dataset contains class imbalance, accuracy alone cannot fully represent classification quality.

B. Precision

Precision measures the proportion of correctly predicted positive samples among all predicted positive samples. It evaluates the reliability of ideological predictions generated by the classifier.

The precision metric is defined as:

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

Higher precision indicates lower false positive predictions and stronger class-specific prediction reliability.

Precision was particularly important for evaluating transformer performance on semantically overlapping ideological categories such as Left versus Leaning-Left and Right versus Leaning-Right.

C. Recall

Recall measures the proportion of correctly identified positive samples among all actual positive samples belonging to a particular class.

The recall metric is defined as:

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

Higher recall indicates stronger capability of the model to identify ideological patterns without missing relevant samples.

Recall was important for evaluating minority ideological category detection performance because imbalanced datasets often cause models to ignore underrepresented classes.

D. F1-Score

The F1-score combines precision and recall into a single metric using harmonic mean calculation. It provides balanced evaluation for datasets containing class imbalance.

The F1-score is defined as:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{4}$$

Since political bias classification involves multiple imbalanced ideological categories, weighted F1-score was selected as one of the primary evaluation metrics during transformer fine-tuning and early stopping.

The weighted F1-score assigns proportional importance to each class based on class frequency, thereby providing more balanced evaluation compared to accuracy alone.

Macro F1-score was also analyzed to evaluate overall class-wise discrimination capability independent of class distribution size.

E. ROC-AUC Score

The Receiver Operating Characteristic - Area Under Curve (ROC-AUC) metric evaluates the model's capability to distinguish between ideological classes across different classification thresholds.

ROC curves visualize the trade-off between:

- True Positive Rate (TPR)
- False Positive Rate (FPR)

A higher ROC-AUC score indicates stronger semantic discrimination capability and better contextual classification performance.

ROC-AUC analysis was particularly useful for evaluating transformer robustness across semantically similar ideological categories.

F. Confusion Matrix

Confusion matrices were used for detailed class-wise performance analysis.

The confusion matrix visualizes:

- Correct classifications
- Misclassifications
- Ideological overlaps
- Semantic confusion patterns

The analysis revealed that most classification errors occurred between neighboring ideological categories such as:

- Left vs Leaning-Left
- Right vs Leaning-Right

This demonstrated the semantic similarity and contextual overlap present within political narratives.

G. Precision-Recall Curve

Precision-Recall curves were analyzed to evaluate transformer performance under class imbalance conditions.

These curves provide deeper understanding of:

- Minority class learning capability
- Prediction confidence quality

- False positive sensitivity

Precision-Recall analysis was especially useful because some ideological classes contained fewer training samples compared to dominant categories.

H. Confidence Distribution Analysis

Confidence distribution analysis was performed to evaluate prediction certainty generated by transformer models.

The confidence distribution plots helped analyze:

- Prediction stability
- Classification confidence
- Uncertain ideological regions
- Semantic ambiguity

This analysis demonstrated that RoBERTa produced more stable and confident ideological predictions compared to lightweight transformer baselines.

I. SHAP-Based Explainability Evaluation

In addition to quantitative metrics, SHAP-based explainability analysis was integrated to evaluate interpretability and semantic feature contribution.

SHAP analysis was used to evaluate:

- Token-level contribution
- Political entity influence
- Contextual semantic importance
- Ideological feature relevance

The explainability evaluation demonstrated that transformer models relied on contextual semantic relationships rather than isolated keyword matching, thereby improving transparency and trustworthiness of ideological classification decisions.

J. Metric Importance for Different Models

Different evaluation metrics were particularly important for different transformer architectures.

- Accuracy was useful for overall comparison between DistilBERT, ALBERT, RoBERTa, and BART + LoRA.
- Weighted F1-score was highly important because the dataset contains imbalanced ideological categories.
- Precision and Recall were useful for analyzing semantic overlap handling capability.
- ROC-AUC evaluated contextual discrimination strength across ideological classes.
- SHAP explainability analysis improved interpretability and transparency of transformer predictions.
- Confidence distribution analysis evaluated prediction stability and semantic certainty.

The combined use of quantitative metrics and explainability analysis provided comprehensive evaluation of political bias classification quality, contextual semantic learning capability, and mitigation-oriented transformer adaptation performance.

VIII. Conclusion and Future Work

A. Conclusion

This work presented an explainable multi-class political bias classification and mitigation framework using transformer-based deep learning models and Explainable Artificial Intelligence (XAI) techniques. The study addressed the problem of identifying ideological bias in political news articles belonging to five categories: Center, Leaning-Left, Left, Leaning-Right, and Right. Multiple transformer architectures, including DistilBERT, ALBERT, RoBERTa, and BART combined with Low-Rank Adaptation (LoRA), were experimentally explored and comparatively evaluated.

The proposed methodology involved dataset preprocessing, tokenization, class balancing, transformer fine-tuning, explainability analysis, and mitigation-oriented adaptation. Among all experimented models, RoBERTa achieved the best overall performance with approximately 76–77% accuracy, a weighted F1-score of nearly 0.77, and strong ROC-AUC performance across ideological categories. SHAP-based explainability analysis further demonstrated that the model relied on contextual political semantics and ideological framing rather than isolated keyword matching. In addition, the BART + LoRA mitigation framework reduced computational complexity and improved parameter-efficient transformer adaptation for scalable deployment.

Overall, the proposed framework successfully combines transformer-based political bias classification, explainable AI analysis, and mitigation-oriented adaptation within a unified and interpretable NLP system. The experimental results demonstrate that contextual transformer learning combined with explainability techniques provides an effective and trustworthy approach for political bias analysis.

B. Future Work

Although the proposed framework achieved strong classification and explainability performance, several improvements and extensions can be explored in future research.

Future work may include:

- Expanding the dataset using multilingual and cross-domain political news sources to improve model generalization capability.
- Exploring larger transformer architectures and modern Large Language Models (LLMs) for deeper contextual understanding.
- Integrating real-time news streaming and online learning mechanisms for dynamic political bias monitoring.
- Improving classification of semantically overlapping ideological categories using hierarchical learning or contrastive learning techniques.
- Extending explainability analysis using advanced visualization frameworks and interactive interpretability methods.
- Exploring additional mitigation strategies such as adversarial debiasing and fairness-aware transformer optimization.

- Applying the proposed framework to related tasks including misinformation detection, propaganda analysis, and media credibility assessment.
- Further optimizing parameter-efficient fine-tuning methods for low-resource and edge-device deployment environments.

The proposed work provides a strong foundation for future research in explainable political NLP systems, trustworthy AI, and bias-aware transformer-based language modeling.

REFERENCES

[1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention Is All You Need," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[2] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in *Proceedings of NAACL-HLT*, 2019.

[3] Y. Liu, M. Ott, N. Goyal, J. Du, *et al.*, "RoBERTa: A Robustly Optimized BERT Pretraining Approach," *arXiv preprint arXiv:1907.11692*, 2019.

[4] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a Distilled Version of BERT: Smaller, Faster, Cheaper and Lighter," *arXiv preprint arXiv:1910.01108*, 2019.

[5] Z. Lan, M. Chen, S. Goodman, K. Gimpel, P. Sharma, and R. Soricut, "ALBERT: A Lite BERT for Self-supervised Learning of Language Representations," in *International Conference on Learning Representations (ICLR)*, 2020.

[6] M. Lewis, Y. Liu, N. Goyal, M. Ghazvininejad, *et al.*, "BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension," in *Proceedings of ACL*, 2020.

[7] E. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, *et al.*, "LoRA: Low-Rank Adaptation of Large Language Models," in *International Conference on Learning Representations (ICLR)*, 2022.

[8] S. Lundberg and S. Lee, "A Unified Approach to Interpreting Model Predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[9] D. M. Blei and J. D. Lafferty, "Topic Models," in *Text Mining: Classification, Clustering, and Applications*, Chapman and Hall/CRC, 2009.

[10] B. Pang, L. Lee, and S. Vaithyanathan, "Thumbs up? Sentiment Classification using Machine Learning Techniques," in *Proceedings of EMNLP*, 2002.

[11] Y. Kim, "Convolutional Neural Networks for Sentence Classification," in *Proceedings of EMNLP*, 2014.

[12] A. Joulin, E. Grave, P. Bojanowski, and T. Mikolov, "Bag of Tricks for Efficient Text Classification," in *Proceedings of EACL*, 2017.

[13] S. Bird, E. Klein, and C. Loper, *Natural Language Processing with Python*, O'Reilly Media, 2009.

[14] Amrita Vishwa Vidyapeetham, "Research Initiatives in Explainable Artificial Intelligence and Trustworthy AI," *Amrita Research Publications*, 2023.

[15] Department of Telecommunications and Amrita Vishwa Vidyapeetham, "AI and Bias Mitigation Research Collaboration for Responsible AI Systems," *Government Research Initiative Report*, 2023.

[16] T. Wolf, L. Debut, V. Sanh, *et al.*, "Transformers: State-of-the-Art Natural Language Processing," in *Proceedings of EMNLP: System Demonstrations*, 2020.

[17] F. Pedregosa, G. Varoquaux, A. Gramfort, *et al.*, "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[18] A. Paszke, S. Gross, F. Massa, *et al.*, "PyTorch: An Imperative Style, High-Performance Deep Learning Library," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[19] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016.

[20] J. Brownlee, *Deep Learning for Natural Language Processing*, Machine Learning Mastery, 2017.